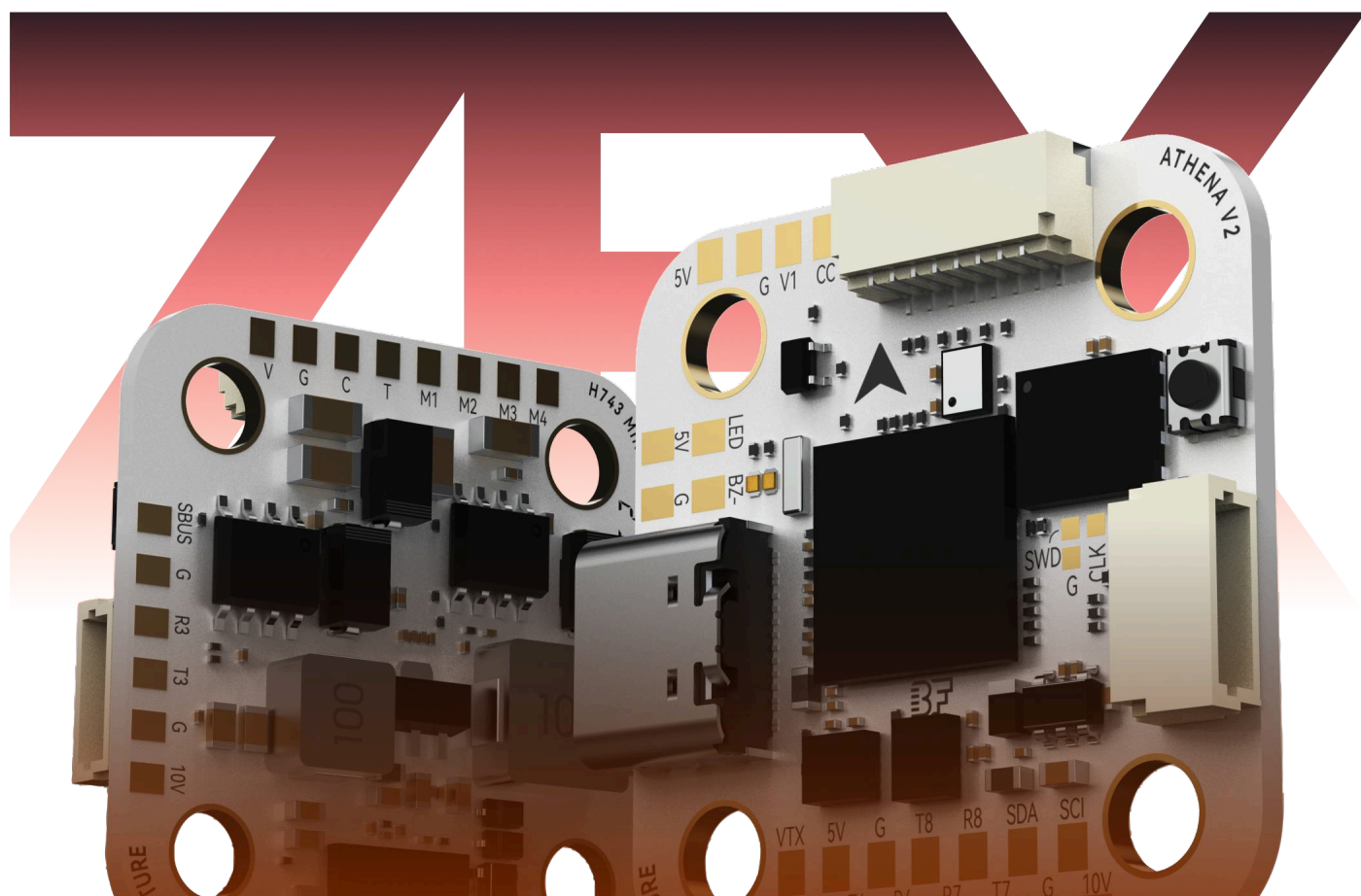




用户手册 泽芯ZEXFPV

ATHENA H743 MINI



如需更多产品信息, 请访问:
www.zexfpv.com

Revision	Date	Description
1.0	26.01.13	Initial Draf

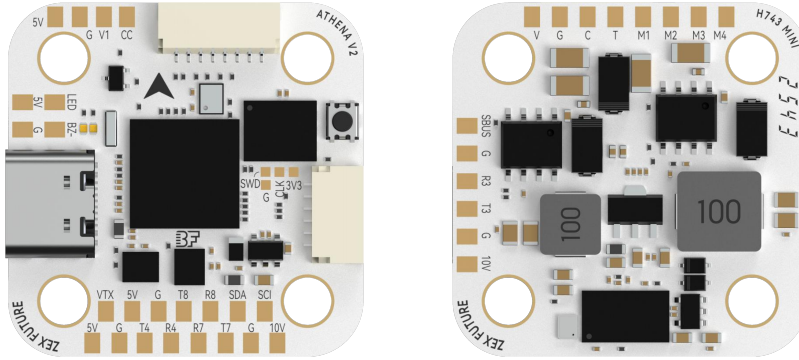
ZEX ATHENA H743 MINI FC

GENERAL

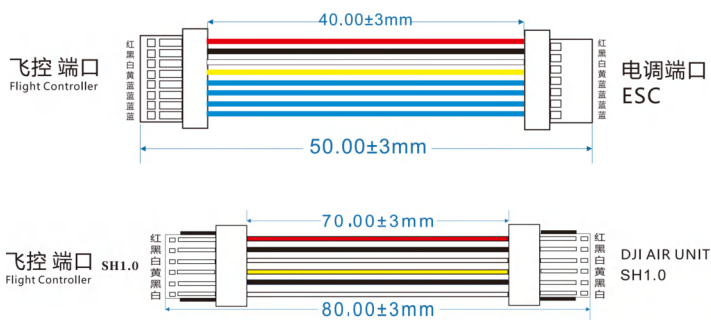
Welcome to the ATHENA H743 MINI FC, a new generation 20mm×20mm flight controller designed by ZEXFPV for those pursuing extreme flight performance and minimal installation space.

We understand that in high-end racing and aerobatics, every gram of weight and every cubic millimetre of space matters. Therefore, the ATHENA H743 MINI FC is not merely a scaled-down version of larger controllers. Instead, it represents a profoundly re-engineered and optimised iteration of the flagship ATHENA H743 MINI FC platform. It inherits all core performance capabilities while achieving revolutionary miniaturisation and integration breakthroughs.

At its heart lies the powerful 480MHz STM32H743 microcontroller, ensuring ample computational headroom. We have equipped it with a premium dual ICM-42688P gyroscope system, alongside our acclaimed independent LDO filtered power supply and master-gyro CLK clock synchronisation technology. This approach significantly suppresses noise at the hardware level, achieving precise alignment between sampling timing and control frequencies. The result is an exceptionally responsive flight experience with unparalleled purity of feel.



ACCESSORIES



1

2

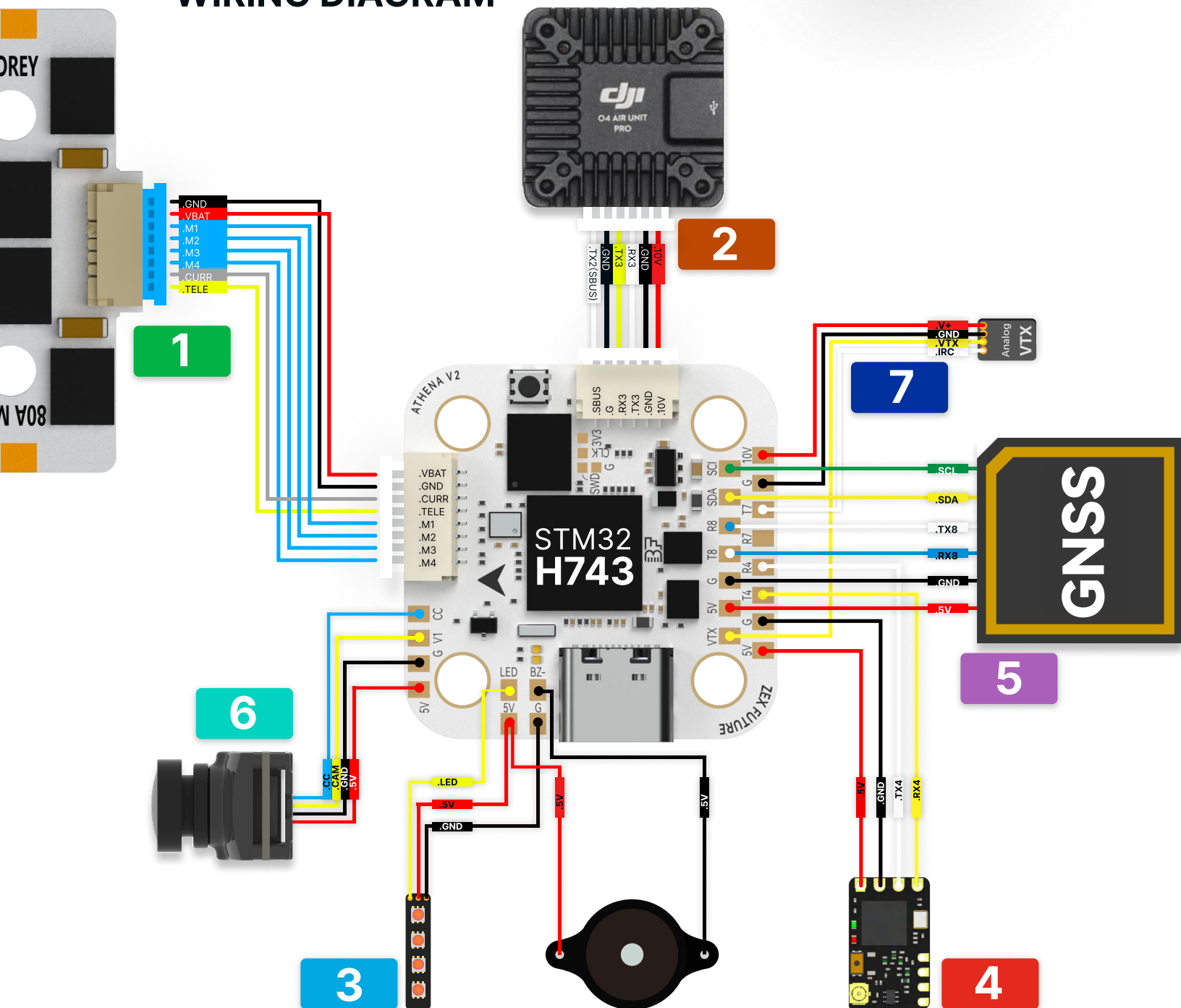


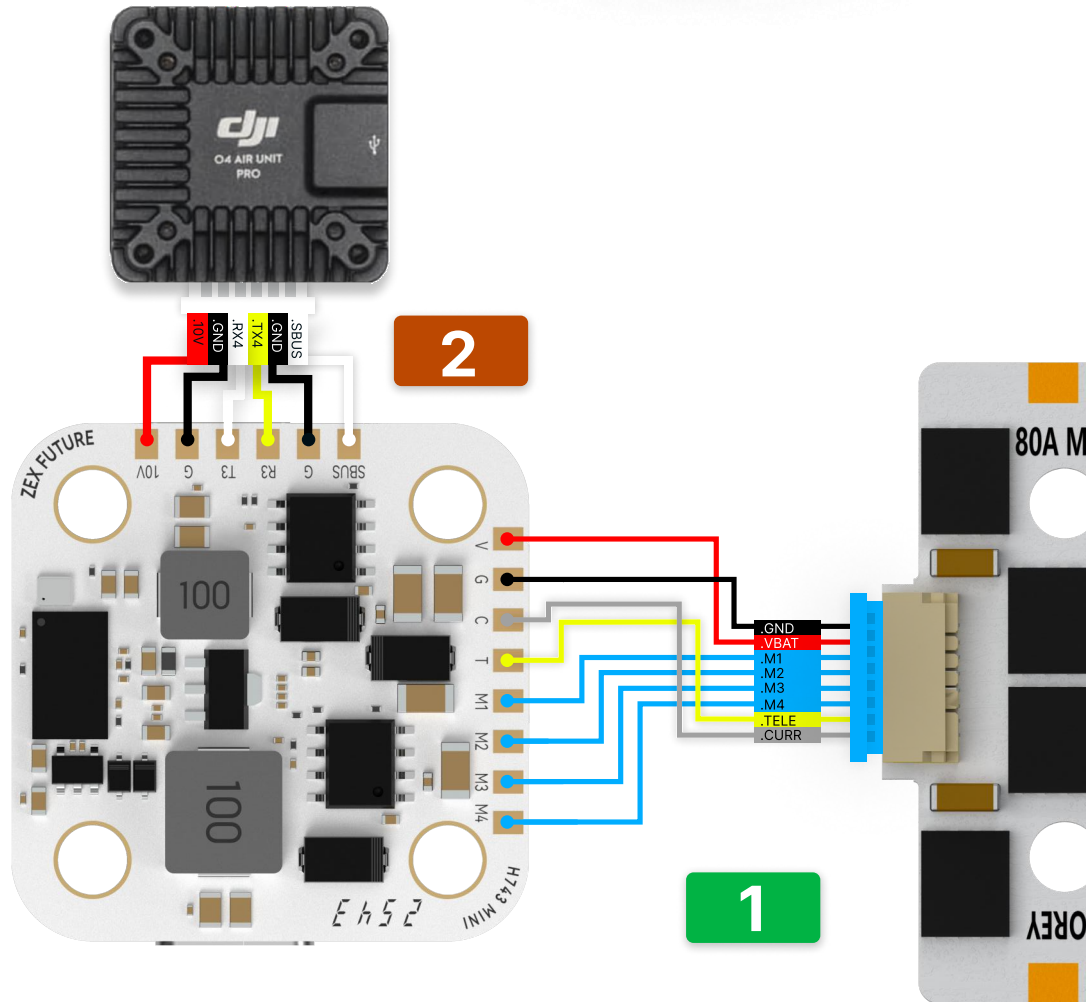
Silicone GEDMMET
M3*6mm x4

SPECIFICATION

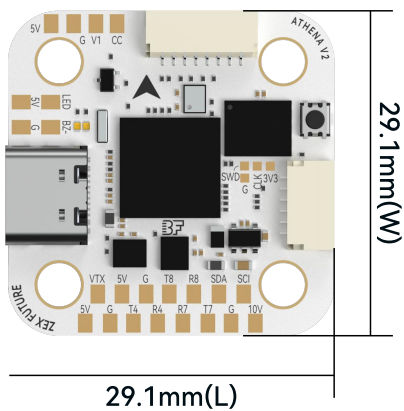
Target	ZEX ATHENA H743 MINI FC
MCU	STM 32H743 (480MHz)
IMU	ICM 42688-P*2
Black Box	windbond 16MB Flash memory
Inner BEC	DC 5V/3A
	DC 10V/2A
4in 1ESC Output	4*Motor Output
USB	TYP-C
UART1	ESC Telemetry
UART2	SBUS
UART3	DJI VTX(Connector)
UART4	RC(FULL UART)
UART7	VTX(FULL UART)
UART8	GNSS(FULL UART)
Anlog OSD	AT7456E
Buzzer	Pads
Mounting	20.5mm*20.5mmxM4
Weight	6.6g
Dimension	29.1(L)* 29.1(W)* 7.6(H)
Barometer	SPA06-003
Battery input voltage	2S-8S

WIRING DIAGRAM





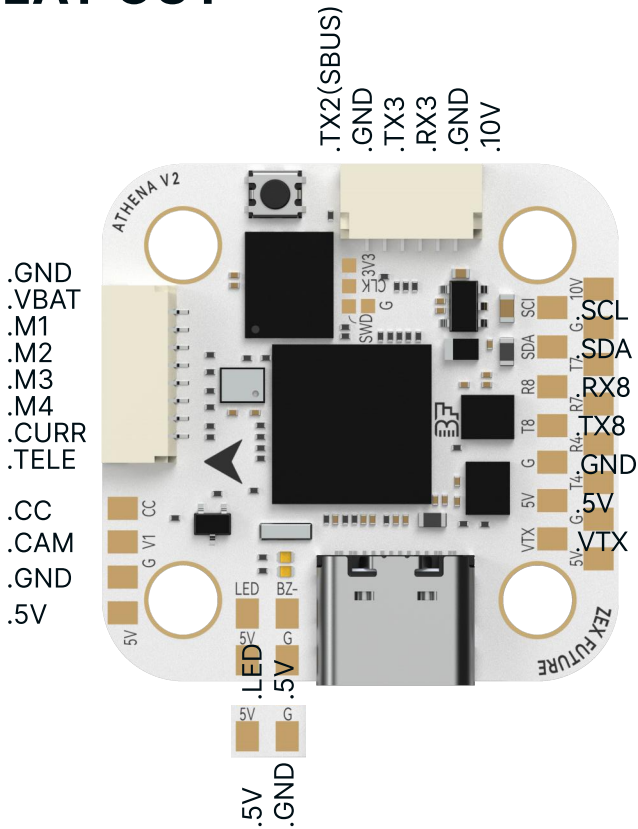
DIMENSION



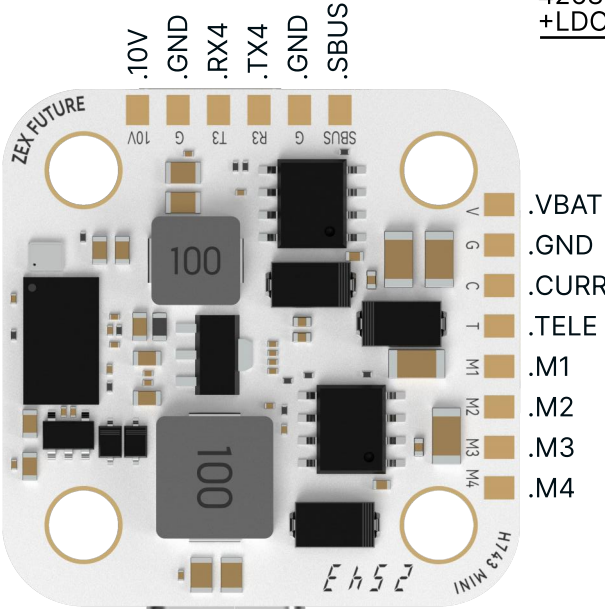
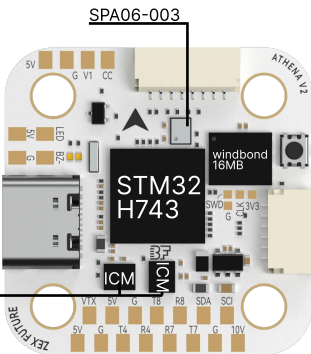
PIN OUT

Pin	Fuction
5V	5V output
G	Ground
VI	Video input from FPV camera
CC	Camera control
LED	Addressable LED signal wire
BZ-	Buzzer negative leg
VTX	Video output to video transmitter
TX8,RX8	UART8 RX and TX
SDA,SCL	I2C connection (for peripherals)
TX4,RX4	UART4 RX and TX
TX7,RX7	UART7 RX and TX
10V	10V output
V (back side)	Vbat Power
G (back side)	Ground
C (back side)	Current
T (back side)	Telemetry
M1 (back side)	Motor 1
M2 (back side)	Motor 2
M3 (back side)	Motor 3
M4 (back side)	Motor 4
SBUS (back side)	For DJI RC
TX3,RX3 (back side)	UART3 RX and TX
10V (back side)	10V output

LAY OUT



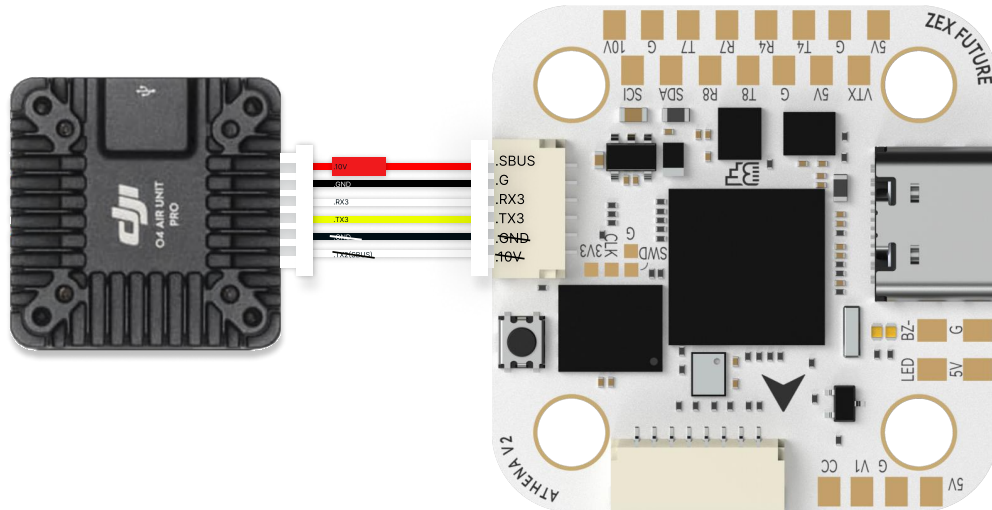
GYRO : ICM
42688-P*2
+LDO filter



SETTINGS

Receiver settings :

If you are not using the DJI Remote Controller, do not connect the SBUS and GND wires.



If using DJI Remote Controller make sure your Receiver Protocol is set to SBUS.

BETAFLIGHT
App Version: 2025.12.1 (4aba25d)
Firmware: 2025.12.1 BTFL
Target: ZEXM/ZEX_ATHENA_MINISTM32H743
2026-01-16 @10:47:41 -- Arming Disabled

0.00V (USB)

Gyro Accel Mag Baro GPS Sonar

Dataflash: free 0B

Update Firmware Disconnect

Show Log

Setup

Ports

Configuration

Power & Battery

Failsafe

Presets

PID Tuning

Receiver

Modes

Adjustments

GPS

Motors

OSD

Video Transmitter

LED Strip

Sensors

Tethered Logging

Blackbox

CLI

Ports

Note: not all combinations are valid. When the flight controller firmware detects this the serial port configuration will be reset.
Note: Do NOT disable MSP on the first serial port unless you know what you are doing. You may have to reflash and erase your configuration if you do.

WARNING: The VTX table has not been set up correctly and without it VTX control will not be possible. Please set up the VTX table in Video Transmitter tab.

Identifier	Configuration/MSP	Serial Rx	Telemetry Output	Sensor Input	Peripherals
USB VCP	115200	☐	Disabled AUTO	Disabled 57600	Disabled 115200
UART1	115200	☐	Disabled AUTO	Disabled 57600	Disabled 115200
UART2	115200	☒ 1	Disabled AUTO	Disabled 57600	Disabled 115200
UART3	115200	☐	Disabled AUTO	Disabled 57600	Disabled 115200
UART4	115200	☐	Disabled AUTO	Disabled 57600	Disabled 115200
UART7	115200	☐	Disabled AUTO	Disabled 57600	Disabled 115200
UART8	115200	☐	Disabled AUTO	Disabled 57600	Disabled 115200

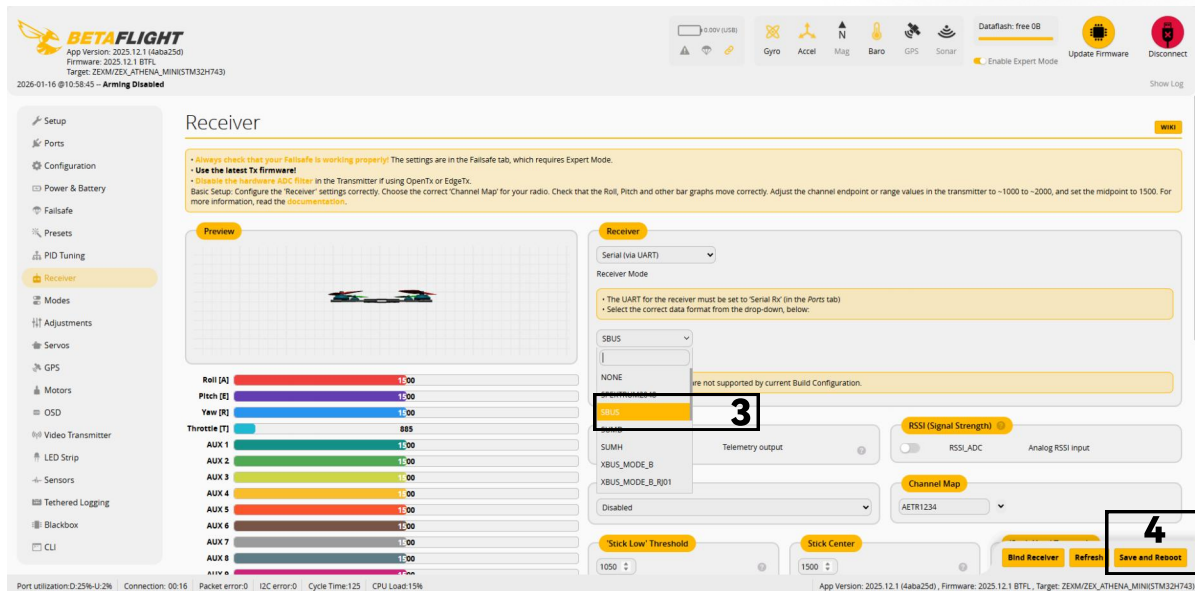
2

Save and Reboot

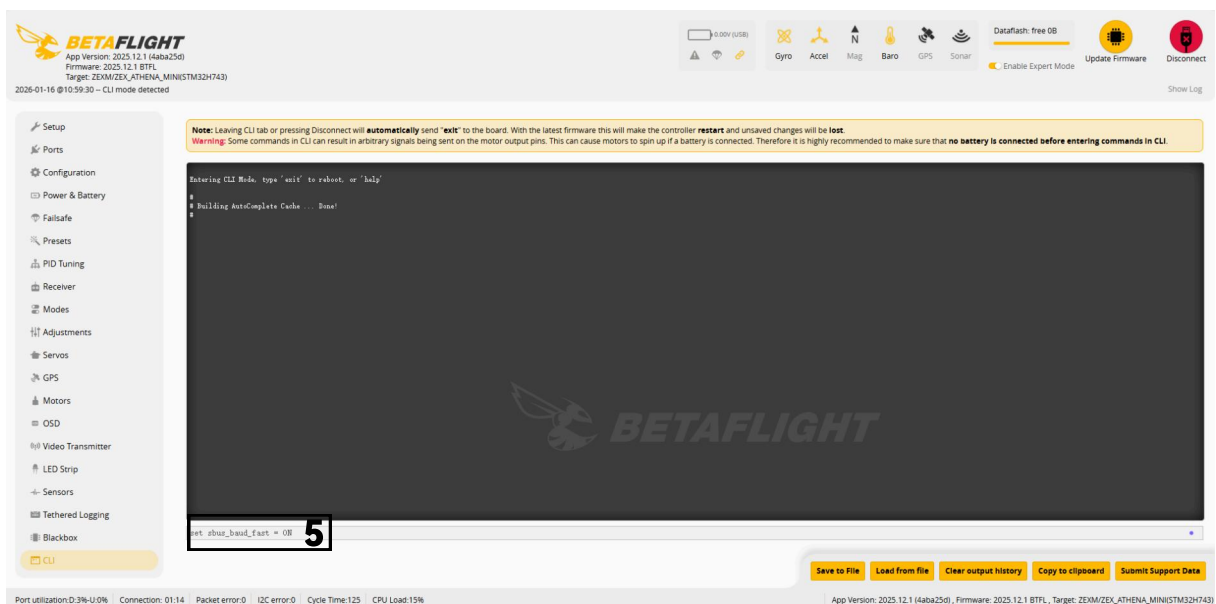
Port utilization: 0.14% - 0.0% Connection: 00:20 Packet error: 0 I2C error: 0 Cycle Time: 125 CPU Load: 15%

App Version: 2025.12.1 (4aba25d), Firmware: 2025.12.1 BTFL, Target: ZEXM/ZEX_ATHENA_MINISTM32H743

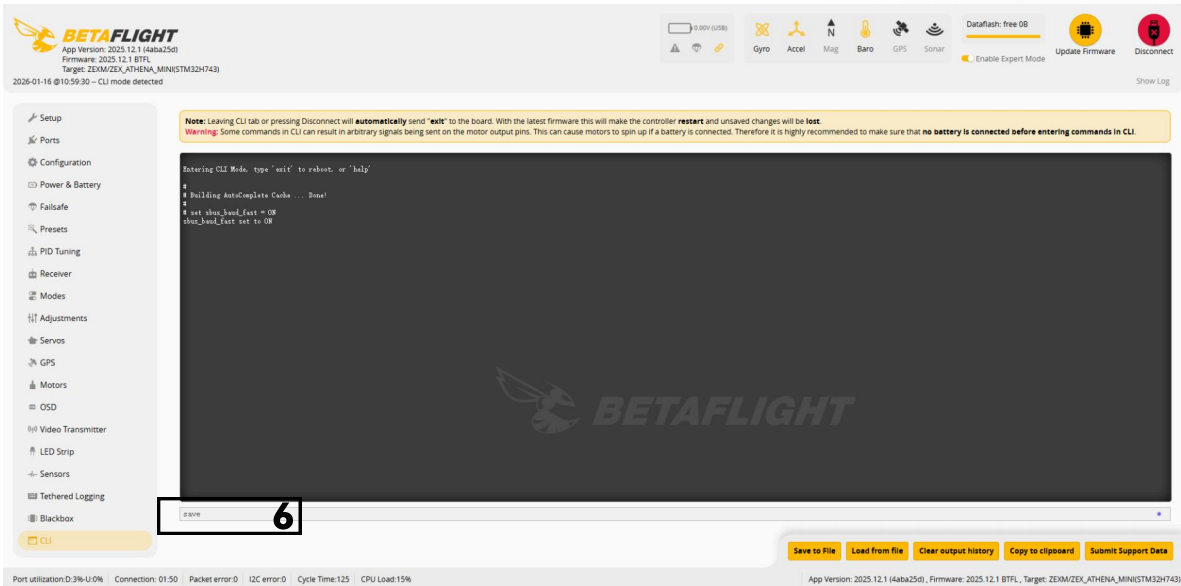
1) Select UART2 as Serial RX, Save and Reboot



2) Select Receiver, set Serial Receiver Provider as SBUS, Save and Reboot



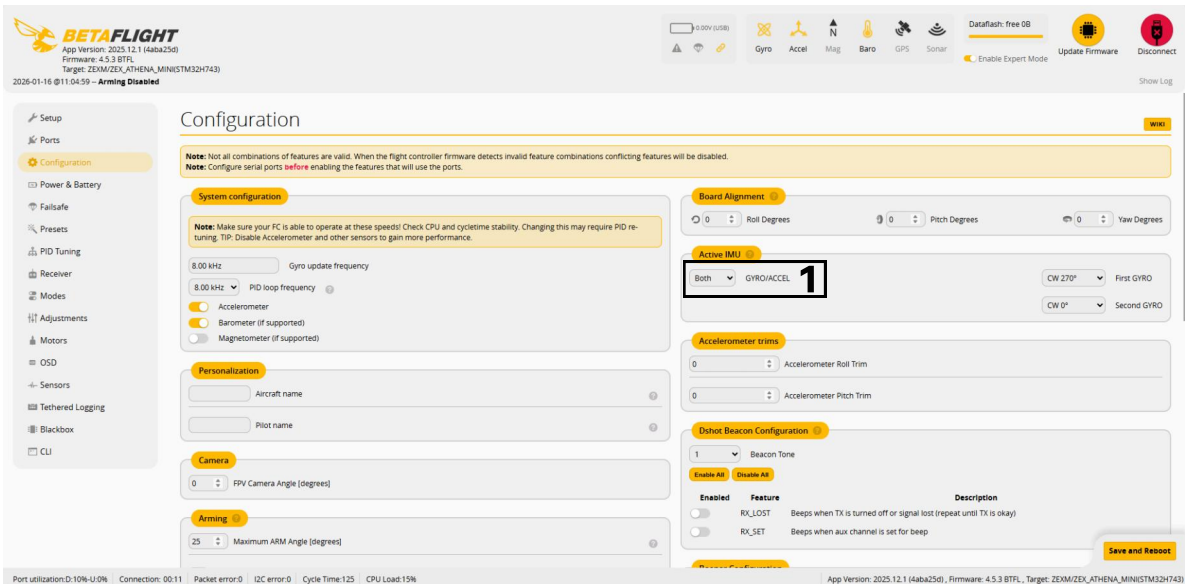
3) Select CLI, Typing set sbus_baud_fast = ON, Enter



4) Typing save, Enter

Gyro Settings :

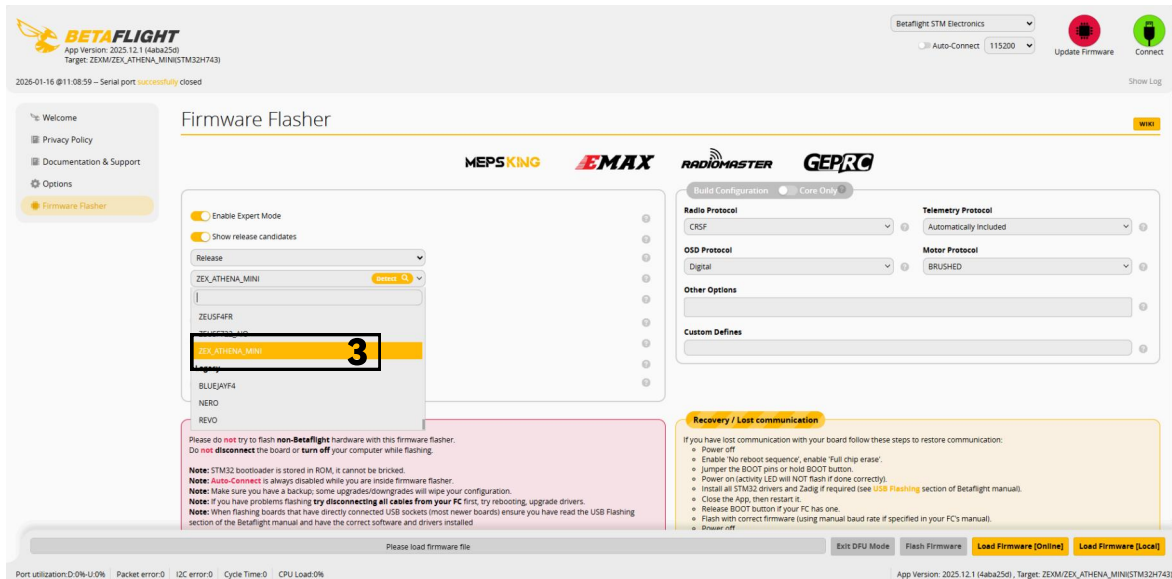
By default, both gyroscopes are enabled simultaneously (the first gyroscope's default orientation is CW 270°, and the second gyroscope's default orientation is CW 0°)



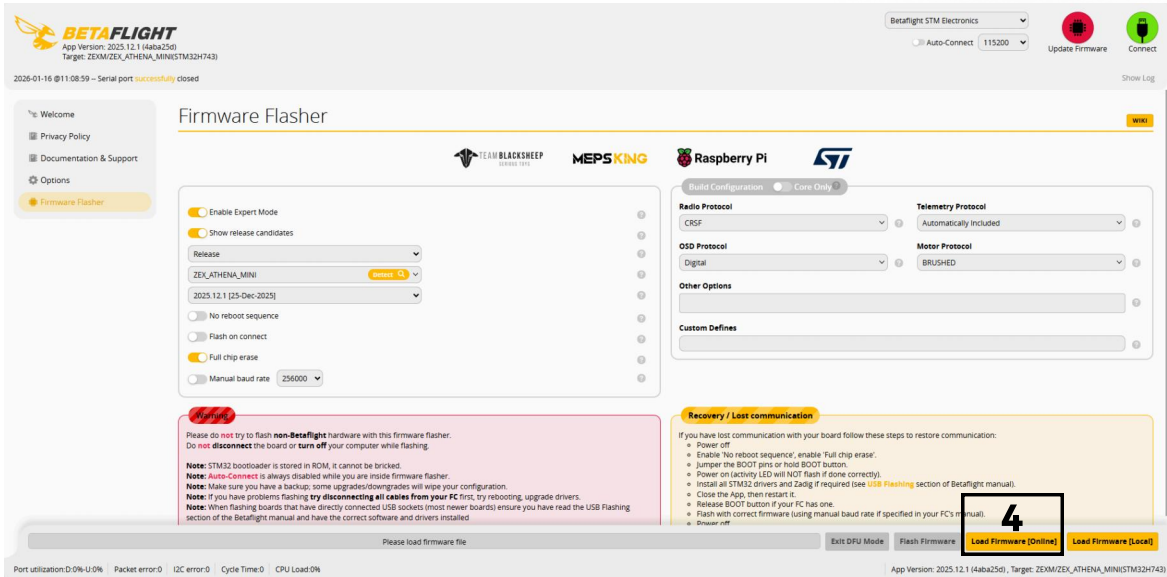
Firmware Target : ZEX_ATHENA_MINI (Betaflight)

Flash Betaflight firmware

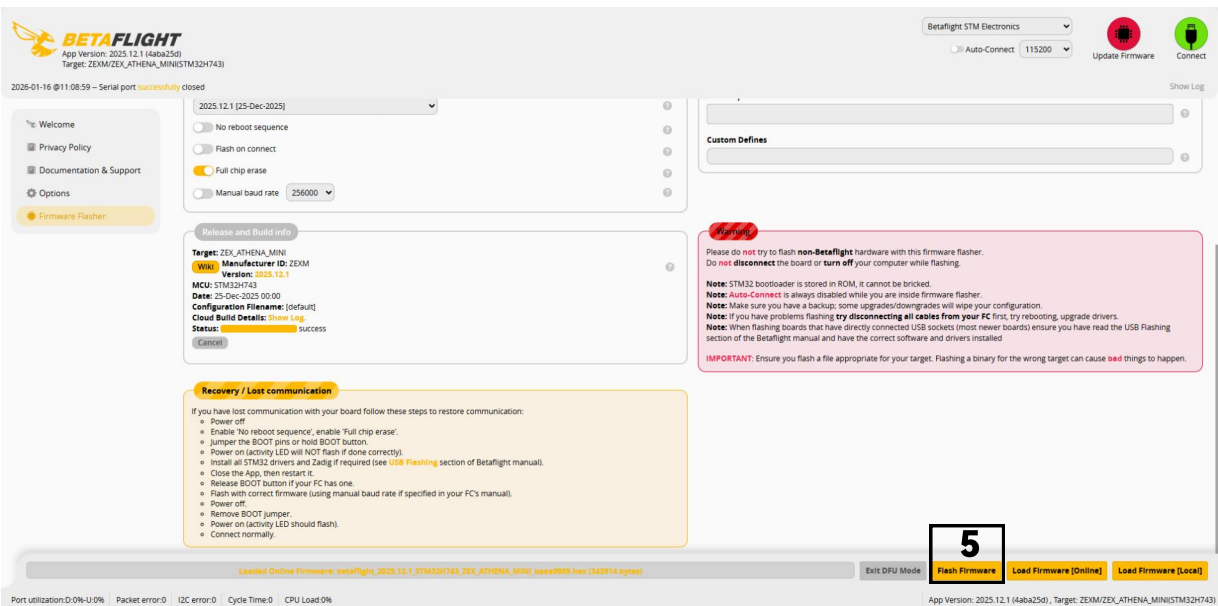
- Download and install the Betaflight Configurator (If flashing firmware versions after 2025.12.1, it is recommended to use the Betaflight website.)
- Launch the Betaflight Configurator
- To flash firmware:
 - 1) Select the target port
 - 2) Click "Update Firmware" to enter Firmware Flasher tab



- 3) Select target "ZEX_ATHENA_MINI" and version, The factory version is 4.5.3[25-Dec-2025]



4) Click "Load Firmware [Online] " to download the firmware



5) Flash Firmware



6) Success